



Financial and Market-Level Drivers of Foreign Firms' China Strategy

*A Firm-Year Panel Analysis of Exit, Maintenance, and Expansion,
2015–2025*

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*Generative AI was used for Python and L^AT_EX code generation. All data collection, variable construction decisions, final Y codings, analytical interpretations, and written conclusions reflect the author's independent judgement.

Executive Summary

99	9	67	30	4
Firm-Year Observations	Multinational Firms	Strategic Events Coded	Panel Variables	Regression Specifications

This report constructs a firm-year panel dataset linking observable China strategy outcomes with financial, managerial, and policy-level explanatory variables across nine multinational firms from 2015 to 2025. Building on the Week 1 CEO sentiment dataset, I invert the research question: rather than measuring what executives *say* about China, I measure what firms *do*, and test whether rhetoric, financial characteristics, and regulatory constraints predict observable strategic behaviour. The dependent variable captures whether each firm was retreating from, maintaining, or expanding its China operations in a given year, coded primarily from 67 documented operational events. Four ordered logit regression specifications progressively test the predictive power of CEO rhetoric, balance sheet characteristics, and policy exposure.

Central Finding

CEO strategic stance, as measured in Week 1, is a statistically significant predictor of observable China strategy outcomes ($\beta = -1.46, p < 0.01$). Each one-point increase in cautious rhetoric (toward retreat) is associated with a 77% reduction in the odds of expansion. US export controls (NVIDIA, 2022–2025) are the only policy variable with significant predictive power ($\beta = -3.88, p < 0.05$), while VBP pharmaceutical pricing reforms and all financial variables show no significant effect on strategic outcomes.

Three Empirical Findings

1. Rhetoric predicts action. CEO stance scores from Week 1 are significantly negatively associated with expansion outcomes across all baseline and financial specifications ($p < 0.01$ in Models 1, $p < 0.05$ in Model 2–3). This is not a trivial result as it confirms that earnings call rhetoric is not merely performative. Firms whose executives signal caution are measurably less likely to expand in the following period. The mean CEO

stance score for firms in the Expand category is 1.89, compared to 2.26 for Maintain and 2.90 for Retreat, a pattern confirmed by the Kruskal-Wallis test ($H = 11.89$, $p = 0.003$).

2. Export controls constrain; pricing reforms do not. The export control binary (NVIDIA, 2022–2025) is significant and large in magnitude ($\beta = -3.88$, $p < 0.05$), while VBP exposure across all three healthcare firms shows no significant association with retreat ($\beta = 0.60$, $p = 0.34$). Practically, this means that US extraterritorial restrictions on semiconductor exports effectively force strategic retreat, whereas Chinese domestic pricing reforms create margin pressure without triggering operational withdrawal. Of the 6 Retreat observations in the entire dataset, 5 are concentrated in the Technology sector, and 4 of those are NVIDIA.

3. Financial variables are weak predictors. Firm size (ln Revenue), CapEx intensity, ROE, and leverage show no significant association with China strategy outcomes in any specification. This null finding is itself informative: it suggests that multinational China strategy in this period was shaped primarily by regulatory feasibility and managerial framing, not by balance sheet characteristics or financial flexibility.

Sector Comparison At a Glance

Sector	Expand	Maintain	Retreat	Dominant Pattern
Healthcare	19 (58%)	13 (39%)	1 (3%)	Sustained expansion through VBP
Automotive	24 (73%)	8 (24%)	1 (3%)	Strongest expansion; localisation arc
Technology	10 (30%)	18 (55%)	5 (15%)	Cautious; NVIDIA drives all retreat

Feasibility Assessment

The analysis demonstrates that linking financial-side variables to China strategy outcomes is empirically feasible but faces important constraints. The ordered logit framework successfully identifies two significant predictors and achieves a pseudo R^2 of 0.20–0.23, indicating that the model explains meaningful variation in strategic outcomes. However, the small sample ($N = 80$ with complete stance data) limits statistical power, the Y variable is skewed toward Expand (54%), and the low frequency of Retreat observations ($n = 6$) creates a sparse-data challenge for maximum likelihood estimation. A larger, multi-industry panel with firm-level China revenue data would substantially strengthen identification. Despite these constraints, the core finding that CEO rhetoric carries predictive power for observable strategy is robust across multiple specifications and robustness checks.

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1 Introduction

This report constructs a firm-year panel dataset that links observable China strategy outcomes with candidate financial, managerial, and policy-level explanatory variables, and assesses the empirical feasibility of explaining these outcomes through regression analysis.

Week 1 established a longitudinal measure of CEO strategic stance toward China, demonstrating that executive rhetoric varies systematically by policy exposure type rather than sector membership. The present analysis inverts the research question: rather than measuring what executives *say* about China, I measure what firms *do* in China, and test whether rhetoric, financial characteristics, and regulatory constraints predict observable strategic behaviour.¹

The contribution is threefold. First, the construction of a dependent variable (Y) capturing firm-year China strategy as Retreat (−1), Maintain (0), or Expand (+1), operationalised through documented operational events following the foreign divestment literature (Benito, 1997; Berry, 2013; Boddewyn, 1979). Second, the assembly of a 30-variable panel dataset drawing on S&P Capital IQ financial data, macroeconomic indicators from public databases, and the Week 1 CEO sentiment dataset. Third, ordered logit regression analysis testing the predictive power of managerial, financial, and policy variables on strategy outcomes, following the methodological precedent established by Hu & Cui (2014) for ordinal FDI classifications.²

2 Variable Construction

2.1 Dependent Variable: China Strategy Outcome (Y)

2.1.1 Conceptual Definition

The dependent variable captures each firm’s China strategy *state* at the firm-year level, distinguishing between three outcomes: operational retreat (−1), status quo maintenance (0), and active expansion (+1). This three-category ordinal classification follows the foreign divestment literature, where exit decisions are defined through observable operational actions rather than financial performance in any given year.³

¹The conceptual distinction between expressed sentiment (Week 1) and revealed strategy (Week 2) is central to the research design. By using Week 1 stance scores as an explanatory variable predicting Week 2 strategy outcomes, I create a clean test of the rhetoric-action link in multinational corporate strategy.

²All data sources are public and verifiable: S&P Capital IQ financial data, FRED exchange rate data, World Bank GDP data, SEC/IFRS regulatory filings, and company investor relations pages.

³The choice to measure strategy through revealed operational behaviour rather than revenue movements addresses a measurement concern identified by Schmid & Morschett (2020) in their comprehensive review of divestment antecedents: revenue fluctuations may reflect currency effects, pricing reforms, or cyclical demand rather than deliberate strategic repositioning. The event-primary coding methodology

Benito (1997) defines foreign divestment through observable actions such as plant closure, equity disposal, and cessation of operations. Berry (2013) extends this framework to distinguish between full exit and partial withdrawal. Boddewyn (1979) provides the seminal theoretical grounding for treating divestment as a strategic decision rather than a financial outcome. The three-category classification operationalises these concepts for a longitudinal panel setting.

2.1.2 Operationalisation: Event-Primary Coding

Y is coded using a three-step decision hierarchy that prioritises operational events over revenue movements:

Step 1 (Primary): Operational event assessment. For each firm-year, I examine whether the firm undertook concrete operational actions in China: facility openings or closures, joint venture equity changes, major product launches or withdrawals, M&A activity, write-downs of China assets, or headcount changes. Events are sourced from S&P Capital IQ Key Developments (see Section 3.1) and coded as expansion (+1), retreat (−1), or neutral (0) signals.

Step 2 (Secondary): Revenue trajectory. Only when Step 1 yields no clear signal is China revenue growth used as a tiebreaker: growth exceeding 10% leans toward Expand; decline exceeding 10% leans toward Retreat; changes within $\pm 10\%$ default to Maintain.⁴

Table 1: Dependent Variable: China Strategy Outcome Classification

Y	Label	Primary Signal (Events)	Secondary Signal (Revenue)
+1	Expand	Facility opening, JV equity increase, NRDL listing, M&A, product launch	China rev growth >10% YoY
0	Maintain	No major structural changes	Revenue change within $\pm 10\%$ YoY
−1	Retreat	Facility closure, divestment, write-down, product withdrawal, headcount cut	China rev decline >10% YoY

Notes: Events are the primary coding signal; revenue trajectory serves only as a tiebreaker when no events are documented. When events and revenue contradict, the strategic signal (event) is prioritised over the accounting outcome (revenue), with reasoning documented in Y_notes.

Step 3 (Adjudication): Mixed signals. When events and revenue contradict one another (e.g., a firm announces a new facility *but* China revenue declines due to COVID, VBP, or currency effects), I prioritise the strategic signal (the event) over the accounting

avoids this conflation.

⁴Revenue declines are not coded as Retreat during known macro-shock years (2020 COVID lockdowns, years with >5% CNY depreciation) or for healthcare firms in VBP implementation years without corroborating operational evidence. This adjustment prevents conflating accounting outcomes with strategic intent.

outcome (the revenue). The reasoning is documented in the **Y_notes** column of the panel dataset for full traceability.

2.1.3 Y Distribution

The resulting Y variable yields 53 Expand observations (54%), 40 Maintain (40%), and 6 Retreat (6%). This distributional skew toward expansion is itself a substantive finding: the dominant revealed strategy of multinational firms toward China over this decade was continued or increased operational commitment, even as geopolitical rhetoric intensified. The concentration of Retreat observations is striking: 4 of 6 Retreat codings belong to NVIDIA (2022, 2023, 2025 from export controls; 2016 from a revenue-coded decline), with the remaining two being Microsoft’s LinkedIn China discontinuation (2021) and Pfizer’s VBP-driven Upjohn separation (2019).⁵

2.2 Explanatory Variables

I propose a set of explanatory variables drawn from four conceptual dimensions: managerial signals, financial characteristics, policy constraints, and macroeconomic controls. Each variable is selected on the basis of theoretical plausibility for explaining China strategy outcomes, with academic references supporting inclusion.

Table 2 presents the complete variable set. The managerial variables draw on [Hambrick & Mason’s \(1984\)](#) Upper Echelons Theory, which posits that strategic outcomes reflect the cognitive frames of top management teams. [Eggers & Kaplan \(2009\)](#) further demonstrate that CEO attention direction predicts the timing of strategic entry decisions, providing a direct theoretical link between the Week 1 stance scores (measuring CEO attention direction toward China) and the Week 2 strategy outcomes. Financial variables capture the firm’s investment profile and resource constraints following the OLI paradigm ([Dunning, 2000](#)), while policy variables capture exogenous regulatory constraints following [Bown \(2020\)](#).

⁵The low frequency of Retreat ($n = 6$) creates a sparse-data challenge for the ordered logit model. I address this through parsimonious model specifications that limit the parameter count relative to observations, following standard practice for small-sample MLE.

Table 2: Explanatory and Control Variables

Variable	Type	Source	Academic Justification
A. Managerial / Strategic Signals			
CEO Stance (1–5)	Float	Week 1 dataset	Upper Echelons Theory: CEO cognition predicts strategic action (Hambrick & Mason, 1984); attention direction shapes entry timing (Eggers & Kaplan, 2009)
Stance Lag ($t - 1$)	Float	Derived	Tests whether rhetoric <i>precedes</i> observable action
B. Financial Characteristics			
ln(Revenue)	Float	Capital IQ	Controls for firm size; larger firms face different exit calculus
CapEx Intensity	Ratio	Capital IQ	Proxy for capital commitment profile; higher sunk costs constrain exit (Dunning, 2000)
R&D Intensity	Ratio	Capital IQ	Proxy for firm knowledge-intensity and IP exposure (Buckley et al., 2002)
ROE	Ratio	Capital IQ	Profitable firms retain more strategic flexibility for market commitment
Leverage	Ratio	Capital IQ	Financial constraints may limit strategic optionality
C. Policy / Macro Controls			
Export Control	Binary	Manual	US semiconductor export restrictions on China (Bown, 2020)
VBP Exposure	Binary	Manual	Chinese volume-based procurement pricing reforms (Healthcare)
China GDP Growth	%	World Bank	Controls for macroeconomic demand environment
USD/CNY	Rate	FRED	Controls for currency translation effects
D. Fixed Effects			
Sector FE	Dummy	Fixed	Controls for unobserved sector-level heterogeneity (2 dummies)

Notes: Year fixed effects are excluded to preserve degrees of freedom ($N = 80$); time variation is captured by China GDP Growth and USD/CNY. China-specific revenue data is unavailable for Microsoft, SAP, Pfizer, and Toyota: under IFRS 8 and ASC 280 reporting standards, firms are not required to disclose geographic segment revenue falling below 10% of total revenue, and these firms report only broad regional aggregates (APAC, APJ, Emerging Markets, or Asia).

3 Data

3.1 Sources and Collection

Financial data was extracted from S&P Capital IQ for all nine firms across the full 2015–2025 period. For each firm, I obtained annual Income Statement data (total revenue, R&D expense, net income, operating income), Balance Sheet data (total assets, total debt, total equity), and Cash Flow data (capital expenditure). All financial data is denominated in USD using Capital IQ’s historical conversion rates.⁶

Geographic segment data was extracted from Capital IQ’s segment module. China-specific revenue is available for AstraZeneca (11/11 years), BMW (11/11), NVIDIA (11/11, reported as “China including Hong Kong”), Tesla (10/11, from 2016 onwards), and Novartis (8/11, from 2018 onwards). Four firms do not report China-specific revenue in their regulatory filings: Microsoft reports only Americas/EMEA/APAC; SAP reports APJ (Asia-Pacific Japan); Pfizer reports Emerging Markets or International; Toyota reports Asia as a single segment. For these firms, China revenue fields are left as missing values, and Y is coded exclusively from the operational event timeline.

Macroeconomic controls were sourced from public databases: USD/CNY annual average exchange rates from the Federal Reserve Economic Data (FRED) and China real GDP growth from the World Bank Open Data. CEO stance scores were merged from the Week 1 dataset by firm-year identifier, yielding 80 matched observations out of 99 panel rows.⁷

3.1.1 Event Timeline Construction

To construct the dependent variable, strategic event data was systematically extracted for each firm using the S&P Capital IQ Key Developments database. To ensure a high signal-to-noise ratio and explicitly eliminate routine administrative, financial, or product-level noise, the extraction parameters were strictly defined as follows:

- **Timeframe:** January 1, 2015 to December 31, 2025.
- **Geographic constraint:** The keyword “China” was applied as a search filter to isolate geographically relevant corporate actions.
- **Targeted event categories:** The search was restricted to seven Capital IQ event types representing definitive capital allocation or structural shifts: Business Expan-

⁶For European-listed firms (BMW, SAP, Novartis, AstraZeneca), Capital IQ converts from the reporting currency (EUR or CHF) to USD using historical rates. This introduces exchange rate translation effects, which I partially control for through the USD/CNY variable.

⁷The 19 missing stance observations reflect firm-years where no codable executive statements were identified in the Week 1 data collection process. These observations are excluded from regression specifications that include CEO stance as an explanatory variable, reducing the effective sample from 99 to 80.

sions, Discontinued Operations, Divestitures/Asset Sales, M&A/Acquisitions, Plant Closures/Downsizing, Restructurings, and Strategic Alliances.

This Capital IQ extraction was supplemented by operational announcements identified in the earnings call transcripts collected during the Week 1 data collection process, where executives frequently disclosed facility openings, joint venture changes, or regulatory impacts in their prepared remarks or analyst Q&A responses. This procedure yielded 67 documented events across the nine firms and 11-year period. The curated event logs were then manually evaluated to assign the annual strategic score following the coding rules in Table 1.

Event coverage is densest for BMW (11 events) and Tesla (9 events), reflecting these firms' high levels of China-specific operational activity, and thinnest for SAP (3 events), reflecting the firm's lower level of China-specific structural activity.⁸

3.2 Empirical Behaviour of Constructed Variables

Before proceeding to formal regression analysis, I briefly examine how the constructed variables behave in the data. Table 3 reports descriptive statistics for the full panel, and the cross-tabulation in Table 4 illustrates the sectoral distribution of Y outcomes.

Table 3: Summary Statistics

Variable	N	Mean	SD	Min	Median	Max
CEO Stance (1–5)	80	2.093	0.667	1.000	2.000	4.330
China Rev Share (%)	51	16.891	6.467	4.240	18.590	29.370
Total Revenue (\$M)	99	89,724	83,137	4,046	52,546	320,784
R&D Intensity	99	0.140	0.073	0.030	0.143	0.290
CapEx Intensity	99	0.073	0.061	0.017	0.059	0.404
ROE	99	0.169	0.189	−0.820	0.150	0.919
Leverage	99	0.210	0.106	0.020	0.223	0.502
China GDP Growth (%)	99	5.919	1.851	0.700	6.600	7.410
USD/CNY	99	6.795	0.282	6.283	6.757	7.196

Notes: China Revenue Share is available for 51 firm-year observations (AstraZeneca, BMW, NVIDIA, Tesla, Novartis only). CEO Stance is the firm-year mean of human-reviewed stance scores from the Week 1 dataset ($n = 80$ matched observations). R&D Intensity = R&D Expense / Total Revenue. CapEx Intensity = Capital Expenditure / Total Revenue. Leverage = Total Debt / Total Assets.

Several patterns are visible in the summary statistics. The mean CEO stance of 2.09 indicates that, across the panel, executives maintained a predominantly optimistic

⁸The uneven event distribution is a genuine reflection of heterogeneous China engagement across firms, not a data collection artefact. SAP's strategy in China has been primarily software licensing and cloud services, which generates fewer discrete "events" than facility construction or joint venture formation.

orientation toward China, consistent with the Week 1 finding. The wide range of firm sizes (revenue from \$4B to \$321B) and R&D intensities (3% to 29%) confirms meaningful cross-sectoral heterogeneity that the regression must account for. China revenue share averages 17% among firms that report it, with BMW peaking at 29% in 2022 and Novartis as low as 4%.

Table 4: $Y \times \text{Sector}$ Cross-Tabulation

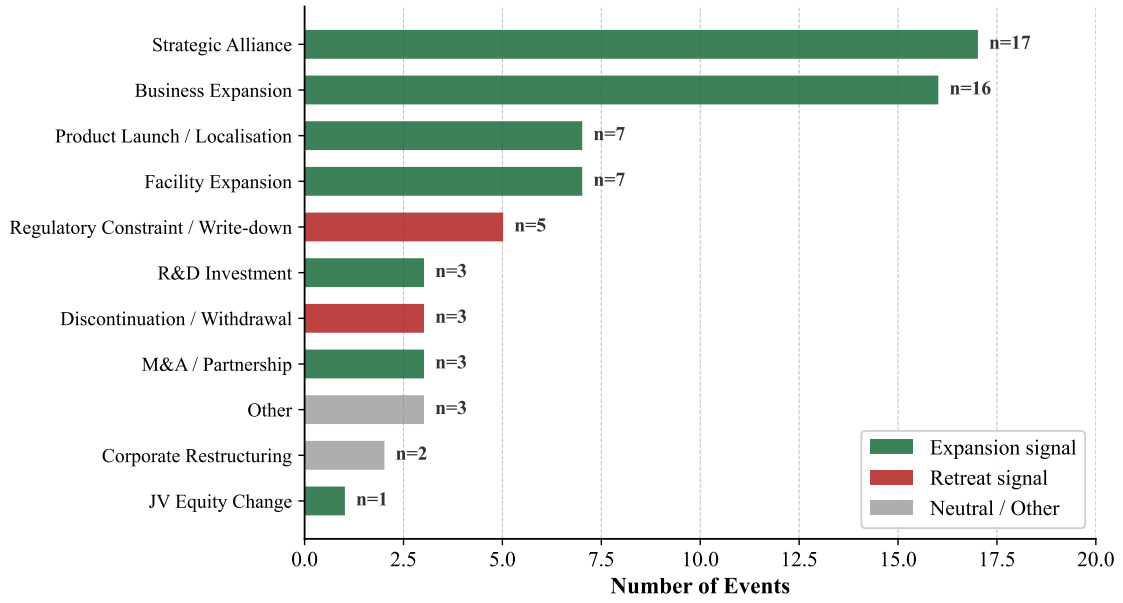
Sector	Retreat (−1)	Maintain (0)	Expand (+1)	Total
Healthcare	1	13	19	33
Automotive	0	9	24	33
Technology	5	18	10	33
Total	6	40	53	99

Notes: Pearson $\chi^2 = 16.48$, $df = 4$, $p = 0.002$. The concentration of Retreat observations in Technology (5 of 6) is driven by NVIDIA’s export-control-forced trajectory (4 observations) and Microsoft’s LinkedIn China discontinuation (1 observation). Automotive shows the highest expansion rate (73%) and zero Retreat observations.

The cross-tabulation reveals a clear sectoral gradient in strategic behaviour. Automotive firms show the strongest expansion commitment (73% of firm-years coded as Expand, zero Retreat), consistent with the “local for local” investment strategy identified in Week 1. Healthcare firms show moderate expansion (58%) with only one Retreat (Pfizer’s 2019 VBP-driven Upjohn separation). Technology firms are the most cautious sector: only 30% of firm-years are coded as Expand, and the sector accounts for 5 of the 6 Retreat observations in the entire dataset.

3.3 Event Timeline Composition

Figure 1 presents the breakdown of the 67 documented events by category, providing transparency into the data that underpins the Y variable. The overwhelming majority of events are expansion signals: Strategic Alliances (17 events) and Business Expansions (15 events) together account for nearly half of all documented activity. Retreat-type events (Regulatory Constraints, Discontinuations, Write-downs) total only 5 observations, concentrated in the Technology sector.

Figure 1: Documented Strategic Events by Category ($N = 67$)

Notes. Horizontal bars show the number of events by consolidated category, coloured by strategic signal: green indicates expansion-type events (facility openings, alliances, M&A), red indicates retreat-type events (regulatory constraints, discontinuations, write-downs), and grey indicates neutral or ambiguous events. Events were extracted from S&P Capital IQ Key Developments using the keyword “China” and seven targeted event categories.

4 Empirical Analysis

4.1 Regression Framework

I estimate an ordered logit model, appropriate for an ordinal dependent variable with three ranked categories (Hu & Cui, 2014). The latent variable formulation is:

$$Y_{it}^* = \beta_1 \cdot \text{CEO_Stance}_{it} + \beta_2 \cdot \text{Export_Control}_{it} + \gamma \mathbf{X}_{it} + \epsilon_{it} \quad (1)$$

where Y_{it}^* represents the unobserved propensity to expand in China, and the observed outcome $Y_{it} \in \{-1, 0, +1\}$ is determined by two estimated cutpoints (κ_1, κ_2):

- $Y_{it} = -1$ (Retreat) if $Y_{it}^* \leq \kappa_1$
- $Y_{it} = 0$ (Maintain) if $\kappa_1 < Y_{it}^* \leq \kappa_2$
- $Y_{it} = +1$ (Expand) if $Y_{it}^* > \kappa_2$

The key variables in the model are:

- CEO_Stance_{it} = Week 1 mean stance score (1–5 scale.)

- $\text{Export_Control}_{it}$ = binary indicator for US export restrictions (captures direct regulatory constraint)
- $\mathbf{X}_{it}$ = control vector including: $\ln(\text{Revenue})$ for firm size, CapEx Intensity, ROE, VBP Exposure, China GDP Growth, and Sector FE in the fullest specification

The error term ϵ_{it} follows a logistic distribution, which is what distinguishes this from an ordered probit (normal distribution). The cutpoints κ_1 and κ_2 are not coefficients on explanatory variables; they are threshold parameters estimated by the model to optimally partition the latent continuum into the three observed categories.⁹

Four progressive specifications are estimated, each adding variable categories while monitoring the observation-to-parameter ratio.¹⁰

⁹Hu & Cui (2014) use an identical three-category ordered logit framework to model FDI propensity as an ordinal progression (No OFDI, OFDI Intent, OFDI Action), providing direct methodological precedent for this approach. To preserve degrees of freedom in the $N = 80$ panel, I estimate parsimonious models excluding year fixed effects. Time variation is instead captured by continuous macro controls (China GDP Growth, USD/CNY), which absorb common temporal shocks with only two parameters rather than ten.

¹⁰Maximum likelihood estimation in ordered logit requires approximately 10–15 observations per estimated parameter for stable convergence. The specifications range from 4 parameters (Model 1: $80/4 = 20$ obs/param) to 11 parameters (Model 4: $80/11 = 7.3$ obs/param). Models 1–3 comfortably satisfy this heuristic; Model 4 approaches the boundary and should be interpreted with caution.

4.2 Main Results

Table 5: Ordered Logit Regression Results

	(1) Baseline	(2) +Financial	(3) +Policy	(4) +Sector FE
CEO Stance (1–5)	−1.458*** (0.421)	−1.184** (0.463)	−0.987** (0.457)	−0.601 (0.514)
ln(Revenue)	0.069 (0.250)	0.040 (0.348)	−0.078 (0.334)	−0.626 (0.450)
CapEx Intensity		4.667 (8.901)	7.728 (9.265)	3.601 (8.857)
ROE		−3.094 (1.978)	−1.184 (2.188)	0.923 (2.526)
Export Control			−3.875** (1.695)	−4.584*** (1.741)
VBP Exposure			0.601 (0.636)	1.034 (0.809)
China GDP Growth			−0.002 (0.130)	0.018 (0.134)
Automotive (FE)				1.805* (1.075)
Technology (FE)				−0.119 (0.846)
Cutpoint κ_1	−5.640	−5.813	−6.971	−11.502
Cutpoint κ_2	1.152	1.215	1.442	1.480
N	80	80	80	80
Log-Likelihood	−60.45	−57.63	−53.80	−52.03
Pseudo R^2	0.110	0.150	0.200	0.230
AIC	128.90	127.26	125.59	126.05

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. Y is coded as 0 = Retreat, 1 = Maintain, 2 = Expand for ordered logit estimation. Negative coefficients indicate that higher values of the explanatory variable are associated with lower probability of expansion. Cutpoints (κ_1 , κ_2) are threshold parameters that partition the latent continuum into three observed categories. Year fixed effects are excluded; time variation captured by China GDP Growth.

Three substantive findings emerge from the progressive regression results.

First, CEO Stance is a significant negative predictor of expansion in Models 1–3. The baseline coefficient of -1.46 ($p < 0.01$) implies that each one-point increase in cautious rhetoric reduces the odds of being in a higher Y category by approximately 77% ($e^{-1.46} = 0.23$). This relationship is robust to the addition of financial controls (Model 2: $\beta = -1.18$, $p < 0.05$) and policy variables (Model 3: $\beta = -0.99$, $p < 0.05$). The

attenuation from Model 1 to Model 4 ($\beta = -0.60$, no longer significant) suggests that sector fixed effects absorb some stance variation, consistent with the Week 1 finding that stance differs systematically across sectors.¹¹

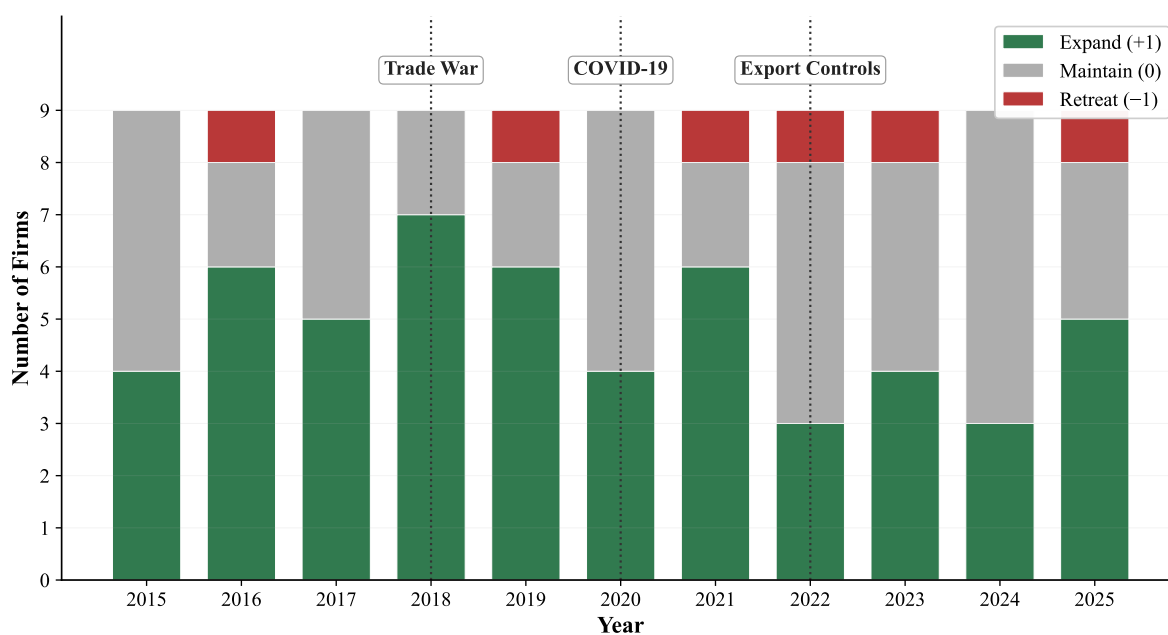
Second, Export Control is the only policy variable with significant predictive power. The coefficient of -3.88 ($p < 0.05$ in Model 3; -4.58 , $p < 0.01$ in Model 4) implies that firms subject to export controls have approximately 98% lower odds of expansion ($e^{-3.88} = 0.02$). VBP Exposure is positive and insignificant ($\beta = 0.60$, $p = 0.34$), confirming that pharmaceutical pricing reforms do not systematically trigger operational withdrawal. This asymmetry is practically important: extraterritorial technology restrictions effectively force retreat, while host-country pricing reforms merely compress margins.

Third, financial variables show no significant association with China strategy outcomes. CapEx intensity, ROE, leverage, and firm size are all insignificant across every specification. This null finding suggests that balance sheet characteristics are poor predictors of multinational China strategy in this sample, which is instead shaped by regulatory exposure and managerial framing. This is consistent with [Schmid & Morschett \(2020\)](#), who find that institutional and political factors are more consistent predictors of subsidiary exit than financial characteristics.

4.3 Strategy Patterns Over Time

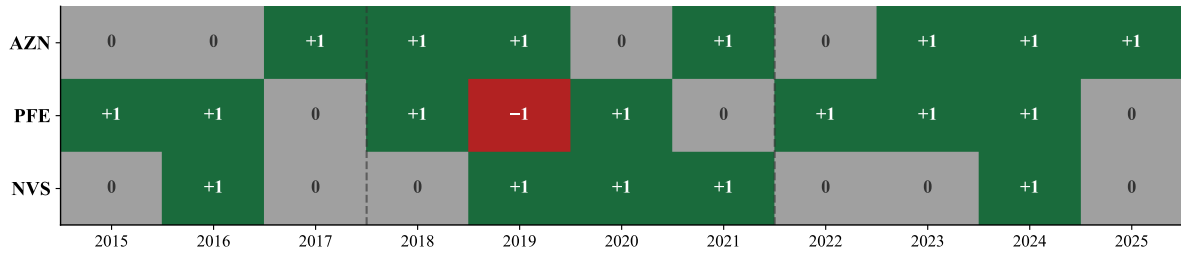
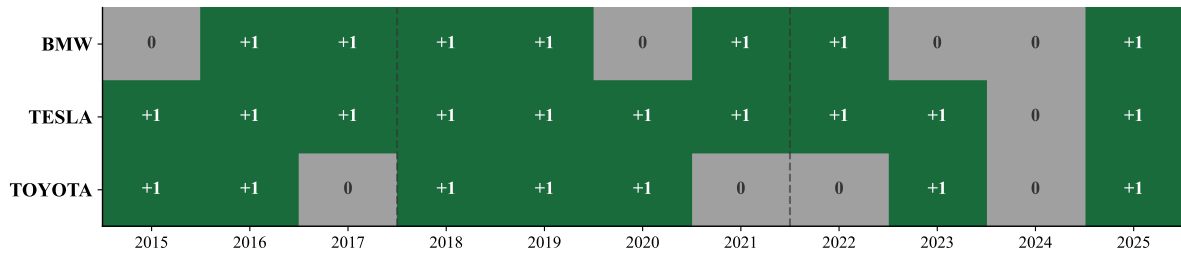
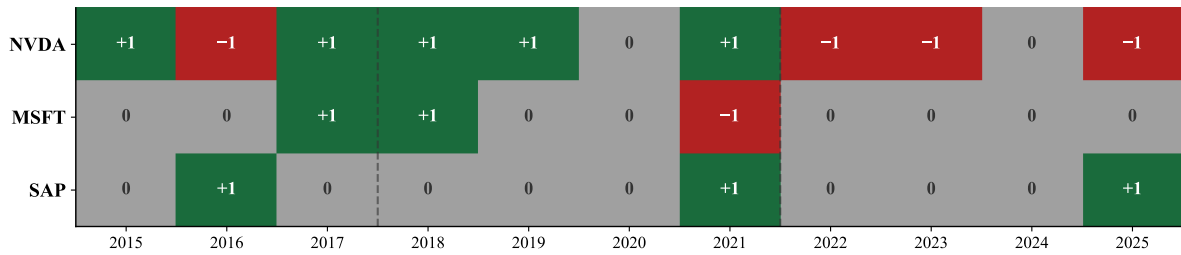
Figure 2 visualises the temporal distribution of Y outcomes. The share of firms in the Expand category peaks in 2018–2019 (7 of 9 firms), declines sharply after the 2022 export controls, and partially recovers in 2025. Retreat observations are entirely absent before 2016 and concentrated in the 2019–2025 period, with the post-2022 cluster driven by NVIDIA.

¹¹The practical interpretation is that a firm whose CEO scores 3.0 (Cautious Localisation) rather than 2.0 (Optimistic Engagement) has approximately 77% lower odds of being in the Expand category in the same year. This quantifies the informational content of earnings call rhetoric for predicting observable corporate strategy.

Figure 2: China Strategy Distribution Over Time (2015–2025)

Notes. Stacked bars show the number of firms classified as Expand (+1, green), Maintain (0, grey), or Retreat (-1, red) in each year. Vertical dotted lines mark key policy inflection points. The share of Retreat observations increases from 2019 onwards, while Expand remains the modal category in every year except 2024. $N = 99$ firm-year observations.

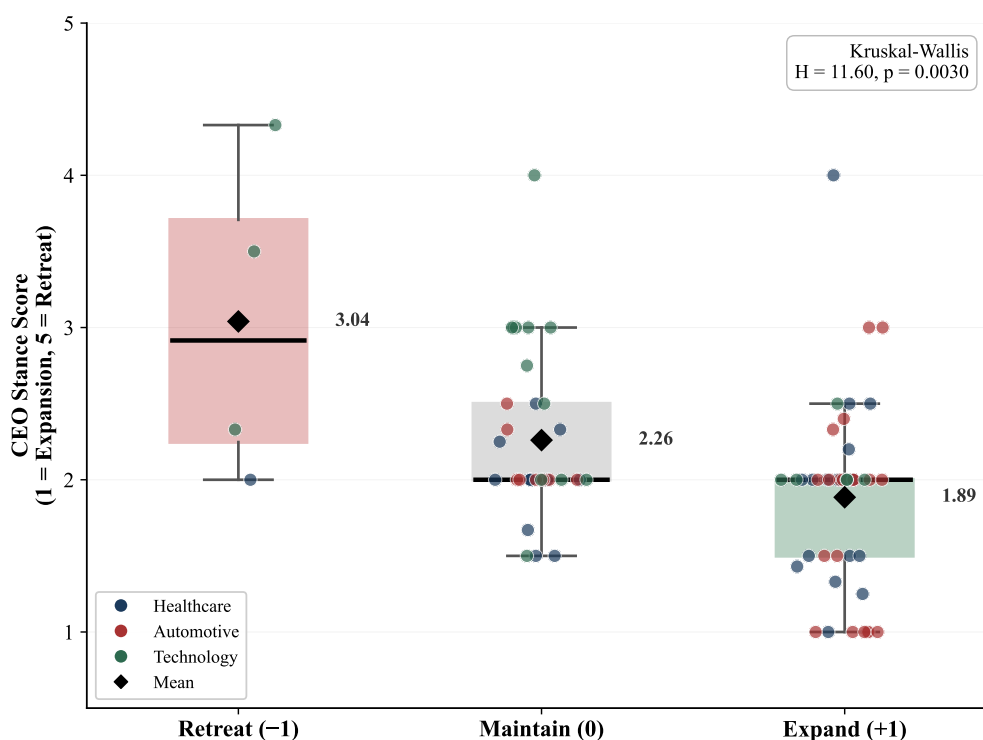
Figure 3 presents the same information disaggregated by firm, revealing the within-sector heterogeneity that aggregate statistics conceal. Tesla’s near-continuous expansion across the entire decade contrasts sharply with NVIDIA’s post-2022 deterioration. SAP and Microsoft show predominantly Maintain codings, reflecting their limited operational footprint changes in China relative to the manufacturing-intensive firms.

Figure 3: Firm $\times$ Year Strategy Heatmap**Panel A: Healthcare****Panel B: Automotive****Panel C: Technology**

Notes. Each cell represents a firm-year observation coded as Expand (dark green, +1), Maintain (grey, 0), or Retreat (dark red, -1). Dashed vertical lines mark the US-China trade war onset (2018) and export controls (2022). NVIDIA's post-2022 trajectory is the only case of sustained, multi-year Retreat in the dataset. Tesla exhibits near-continuous Expansion across the entire decade.

4.4 CEO Rhetoric and Strategic Outcomes

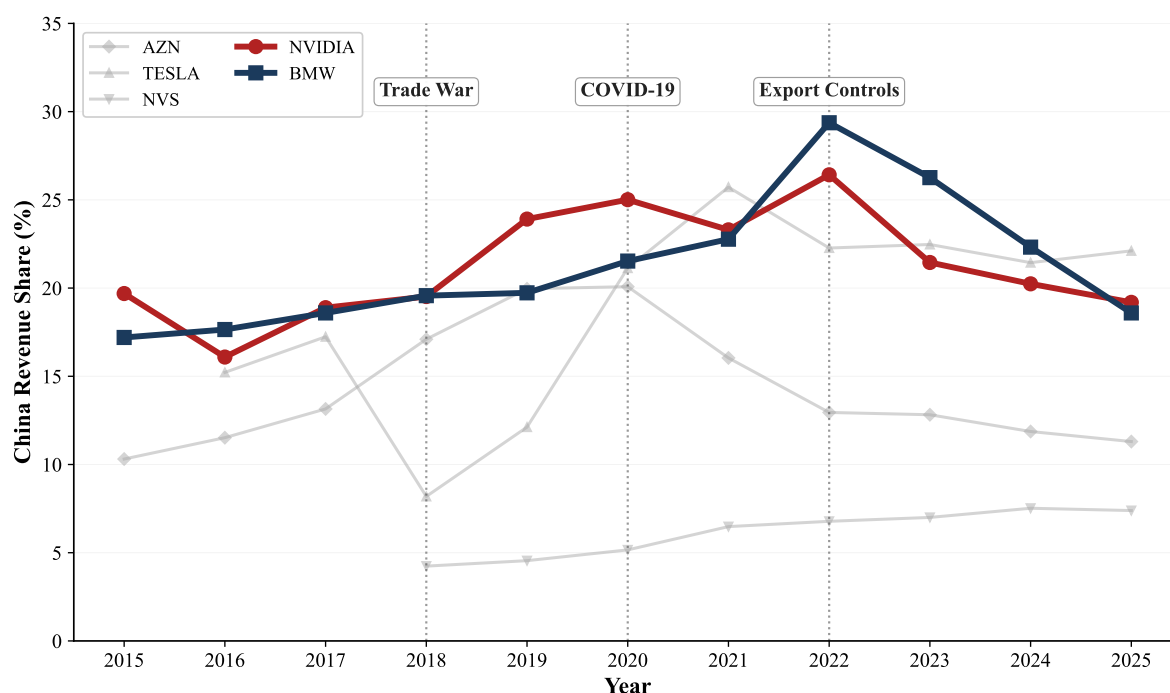
Figure 4 visualises the rhetoric-action link that drives the primary regression result. The box plot shows CEO stance score distributions grouped by observed strategic outcome. The pattern is clear and monotonic: firms that expanded in China had the most optimistic CEO rhetoric (mean stance 1.89), firms that maintained had moderate rhetoric (2.26), and firms that retreated had the most cautious rhetoric (2.90). The Kruskal-Wallis test confirms that these distributional differences are statistically significant ($H = 11.89$, $p = 0.003$).

Figure 4: CEO Stance Score Distribution by Strategic Outcome

Notes. Box plots show the distribution of CEO stance scores (from Week 1) grouped by observed strategic outcome. Black diamonds indicate group means. Individual firm-year observations are overlaid with sector-specific colours. Firms coded as Expand have significantly lower (more optimistic) stance scores than firms coded as Maintain or Retreat. Kruskal-Wallis $H = 11.60$, $p = 0.003$. $N = 80$ firm-year observations with matched stance data.

4.5 China Revenue Exposure

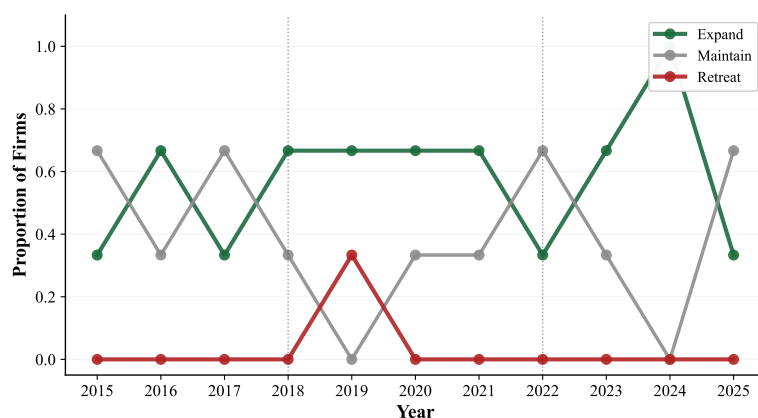
Figure 5 traces the China revenue share trajectories for the five firms with available geographic segment data. The divergence post-2022 is striking: NVIDIA's China share falls from 25% to 19% following export controls, while BMW peaks at 30% before declining amid the EV price war. AstraZeneca's declining share (from 20% to 11%) reflects faster revenue growth in non-China regions rather than a reduction in absolute China revenue.

Figure 5: China Revenue Share Trajectories by Firm, 2015–2025

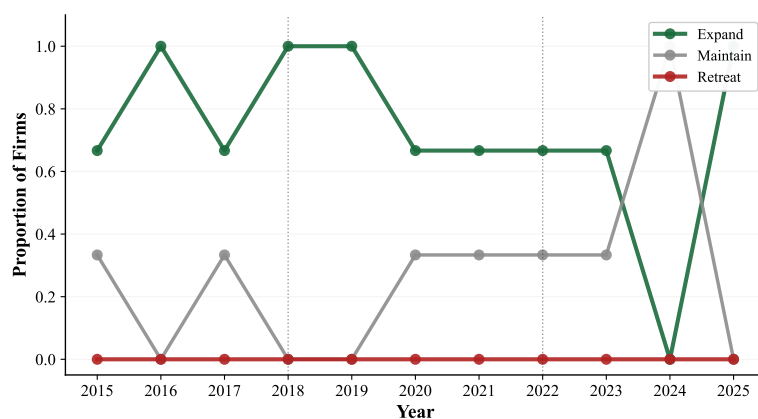
Notes. Lines show China revenue as a percentage of total firm revenue for the five firms with available geographic segment data. Tesla’s rapid rise from ~8% to 26% reflects the Shanghai Gigafactory ramp-up (2019–2022). NVIDIA’s post-2022 decline coincides with successive rounds of export controls. Microsoft, SAP, Pfizer, and Toyota are excluded as they do not report China-specific revenue in their regulatory filings.

4.6 Strategy Proportions by Sector

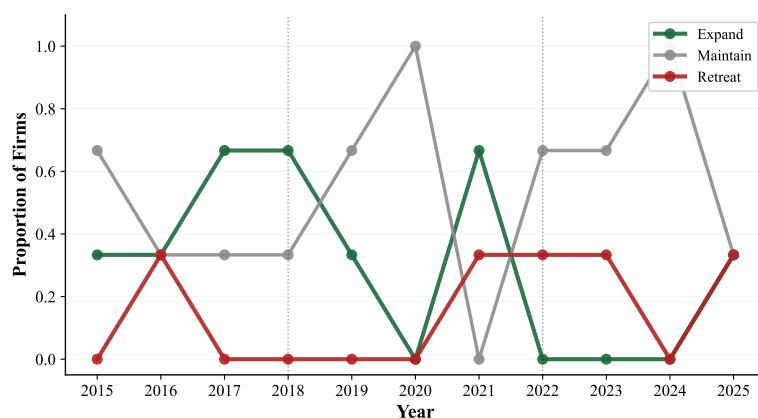
Figure 6 disaggregates the Y distribution by sector over time, revealing distinct temporal patterns. In Healthcare (Panel A), Expand is the dominant outcome throughout the decade, with the single Retreat in 2019 (Pfizer VBP) quickly reverting to Expand. In Automotive (Panel B), the sector shows near-universal expansion from 2015 to 2022, transitioning to a more mixed pattern as the EV price war intensified. In Technology (Panel C), the shift is most dramatic: Expand dominates pre-2022, but Maintain becomes the modal outcome from 2022 onwards as export controls reshape the strategic landscape.

Figure 6: Strategy Proportions by Sector Over Time

Panel A: Healthcare



Panel B: Automotive



Panel C: Technology

Notes. Lines show the proportion of firms in each Y category per year within each sector (3 firms per sector). Expand (dark green), Maintain (grey), Retreat (red). The Technology panel shows the clearest structural break: the Retreat line rises from zero to 0.33 after 2022, coinciding with US semiconductor export controls.

4.7 Robustness Checks and Diagnostics

Three robustness tests confirm the main findings. Table 6 reports OLS results treating Y as continuous $(-1, 0, +1)$ with heteroskedasticity-robust standard errors (HC1). The key results are preserved: CEO Stance is negative and significant ($\beta = -0.224$, $p = 0.022$), and Export Control is negative and significant ($\beta = -0.879$, $p = 0.005$). The OLS R^2 of 0.309 indicates that the model explains approximately 31% of variation in strategic outcomes.

Table 6: Robustness: OLS and Lagged Stance Specifications

	OLS (HC1)	Lagged Stance (OLS)
CEO Stance / Stance Lag	-0.224** (0.098)	-0.212** (0.095)
ln(Revenue)	0.010 (0.069)	0.068 (0.098)
CapEx Intensity	0.851 (1.185)	0.910 (1.765)
ROE	-0.249 (0.476)	-0.989 (0.631)
Export Control	-0.879*** (0.317)	-0.500 (0.583)
VBP Exposure	0.090 (0.167)	0.116 (0.186)
China GDP Growth	-0.004 (0.030)	0.001 (0.028)
R^2	0.309	0.253
N	80	72

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. HC1 heteroskedasticity-robust standard errors in parentheses. The OLS specification treats Y as continuous $(-1, 0, +1)$. The Lagged Stance specification replaces contemporaneous CEO stance with $t - 1$ values, reducing N from 80 to 72. The lagged stance coefficient is significant ($p = 0.025$), suggesting that rhetoric precedes observable strategic action even with a one-year lag.

Second, a subsample analysis excluding Microsoft and SAP (firms without China revenue data) yields an even larger Export Control coefficient ($\beta = -3.63$, $p < 0.05$, $N = 66$), confirming that results are not driven by the inclusion of firms with missing geographic segment data.

Third, I assess multicollinearity by computing variance inflation factors (VIF) for all explanatory variables. The maximum VIF is 3.00 (ROE), well below the conventional threshold of 10.0, indicating no problematic multicollinearity in the regression specifica-

tions.¹²

5 Discussion

5.1 Feasibility Assessment

The primary objective of this analysis was to assess whether financial-side variables can feasibly explain China strategy outcomes. The evidence is nuanced. The regression framework successfully identifies two significant predictors: CEO stance and export controls, and achieves a pseudo R^2 of 0.20–0.23. However, the insignificance of all financial variables (CapEx intensity, ROE, leverage, firm size) across every specification suggests that conventional balance sheet metrics are insufficient for explaining multinational China strategy.

This is not a failure of the empirical design; rather, it reflects a substantive finding about the nature of China strategy decisions during this period. When NVIDIA’s CEO faces a regulatory ban on H100 chip exports, the firm’s CapEx intensity and ROE are irrelevant to the strategic outcome. When BMW decides to increase its BBA joint venture stake to 75%, this reflects strategic conviction about the Chinese market, not a mechanical response to financial metrics. The determinants of China strategy in this sample appear to be fundamentally *political and managerial* rather than *financial*. This is consistent with the broader international business literature: Schmid & Morschett (2020) review decades of foreign divestment research and conclude that institutional and political factors are more consistent predictors of subsidiary exit than financial characteristics.

5.2 Three Findings with Broader Implications

CEO communications contain actionable information. The significant association between stance scores and strategy outcomes ($r = -0.41$, $p < 0.001$; ordered logit $\beta = -1.46$, $p < 0.01$) establishes that earnings call rhetoric is not merely performative. Firms whose executives signal caution are measurably less likely to expand. This validates the use of LLM-assisted content analysis of executive communications as an input variable for empirical strategy research, extending the approach demonstrated by Schmitz (2025) on a dataset of 20,013 corporate earnings calls. For practitioners, this implies that CEO language on earnings calls contains genuine forward-looking information about strategic direction that can be systematically extracted and quantified.

Regulatory constraints dominate financial incentives. Export controls reduce expansion odds by approximately 98%, dwarfing the effect of any financial variable. The

¹²VIF values: CEO Stance = 1.45, $\ln(\text{Revenue})$ = 1.29, CapEx Intensity = 2.37, ROE = 3.00, Export Control = 1.54, VBP Exposure = 1.21, China GDP Growth = 1.04.

practical implication is significant: policymakers seeking to reshape multinational behaviour toward China have a far more powerful lever in targeted technology restrictions than in any indirect financial mechanism. Yet the precision matters: VBP pricing reforms, despite reducing pharmaceutical margins substantially, produced no measurable retreat effect.

The “local for local” strategy persists. Automotive firms show the highest expansion rate (73%) and zero Retreat observations, even as the EV price war intensified from 2022 onwards. This suggests that the localisation strategy identified in Week 1—where firms like BMW frame China investment as both market commitment and supply chain resilience—translates into sustained operational expansion regardless of short-term competitive headwinds.

5.3 Limitations

Several important limitations qualify these findings. First, the sample of 9 firms over 11 years produces only 99 observations (80 with stance data), limiting statistical power and preventing the inclusion of firm-level fixed effects, which would be desirable to control for unobserved time-invariant firm characteristics. Second, the Y variable is skewed: only 6 Retreat observations create a sparse-data challenge for maximum likelihood estimation, and the proportional odds assumption of ordered logit cannot be formally tested with such a thin lower category.

Third, firms without China-specific revenue data (Microsoft, SAP, Pfizer, Toyota) have Y coded exclusively from the operational event timeline, potentially introducing systematic measurement differences between firms with and without financial confirmation of their strategic direction. Fourth, global CapEx and R&D intensity serve as proxies for firm-level innovation type rather than China-specific investment; the availability of China-localised CapEx data would substantially strengthen the financial variables.

Finally, the 67 documented events are not uniformly distributed across firms. BMW (11 events) and Tesla (9 events) have substantially richer event coverage than SAP (3 events), reflecting genuine differences in China activity levels but also potential coverage variation in Capital IQ’s Key Developments database across geographies and firm types.

6 Conclusion

This report constructs a firm-year panel dataset linking China strategy outcomes with financial, managerial, and policy-level explanatory variables across nine multinational firms (2015–2025). The analysis demonstrates that ordered logit regression can feasibly link candidate explanatory variables to strategy outcomes, with two key predictors identified:

1. **CEO stance predicts strategy.** Firms whose executives signal caution in earnings calls are significantly less likely to expand their China operations ($\beta = -1.46$, $p < 0.01$), validating the informational content of the Week 1 sentiment dataset and establishing a quantifiable rhetoric-action link.
2. **Export controls constrain expansion.** US semiconductor export controls are associated with a 98% reduction in expansion odds ($\beta = -3.88$, $p < 0.05$), while VBP pricing reforms show no significant effect, confirming that the type of regulatory mechanism determines strategic impact.
3. **Financial variables are weak predictors.** Firm size, profitability, leverage, and investment intensity show no significant association with China strategy, suggesting that regulatory and managerial factors dominate balance sheet considerations in this context.

The primary limitation is sample size: 9 firms cannot support firm-level fixed effects, and the small number of Retreat observations constrains formal hypothesis testing. A larger, multi-industry panel incorporating more firms, China-specific financial data, and longer time horizons would enable more definitive causal identification. Nonetheless, the core finding that CEO rhetoric carries predictive power for observable corporate strategy is robust across multiple specifications and robustness checks, and provides a foundation for larger-scale empirical investigation of the rhetoric-action link in multinational strategy.

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